

Function vectors.

↳ characterize "how ICL works."
↓
In-context learning.

ICL $\xrightarrow{?}$ Abstracting out the
"task"

Hidden vectors in transformer → affected.

↳ obtain a compact vector
representation of the emulated
the process used by the
ICL

transformer $\rightarrow$ solve
a/multiple tasks.

predicting unknowns.

FVs are
used to solve
simple (few
tokens) tasks.

$T_x \rightarrow$ ICL prompt $\rightarrow$ demonstrate a task t

Q1: a) Gather a set of ICL prompts for the P_t
task t

⑥ Avg activation $\bar{h}_l^t$ (avg activation for the
task t in layer l).
at the last token of each prompt
(in each layer).

③ Intervention: Add h_l to the representation after l when the transformer completes a zero-shot task

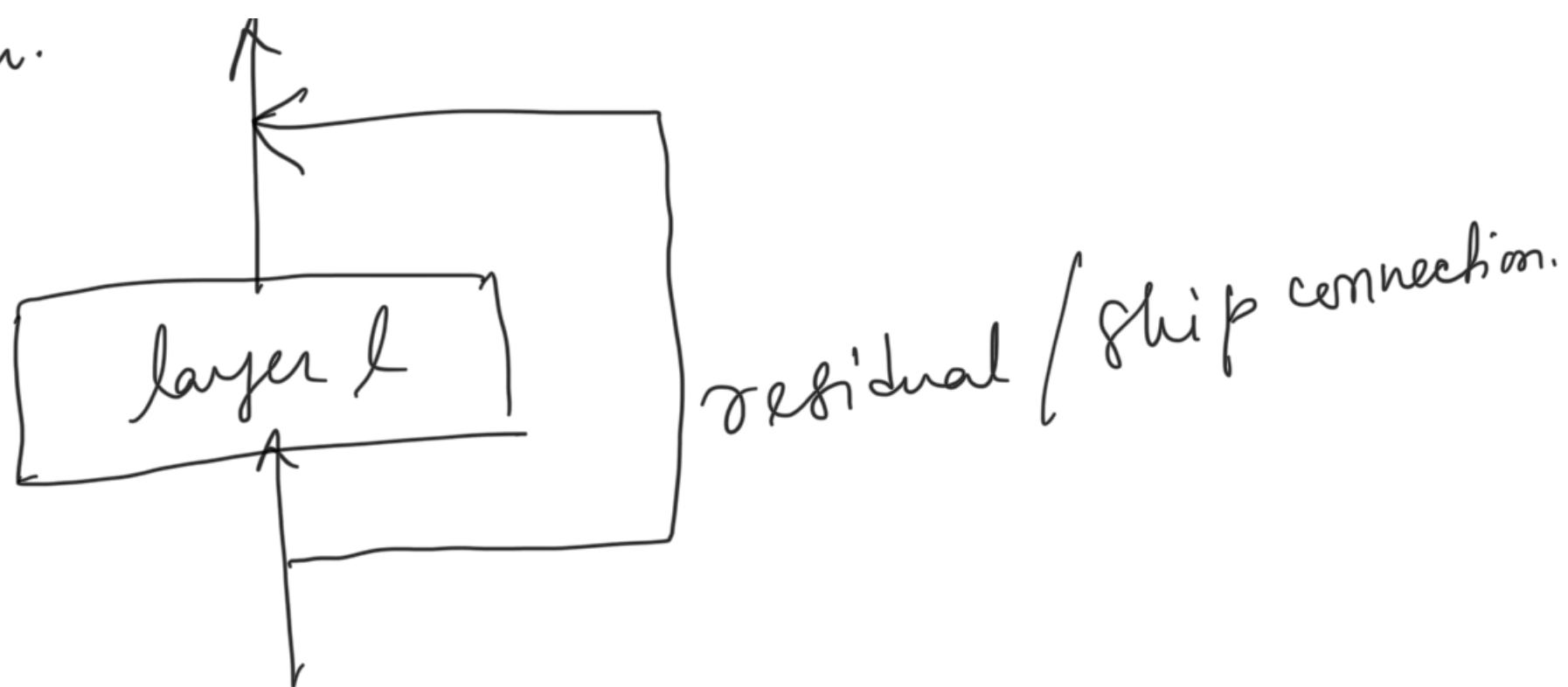
→ solve a task in a new context!

t = antonym prediction.

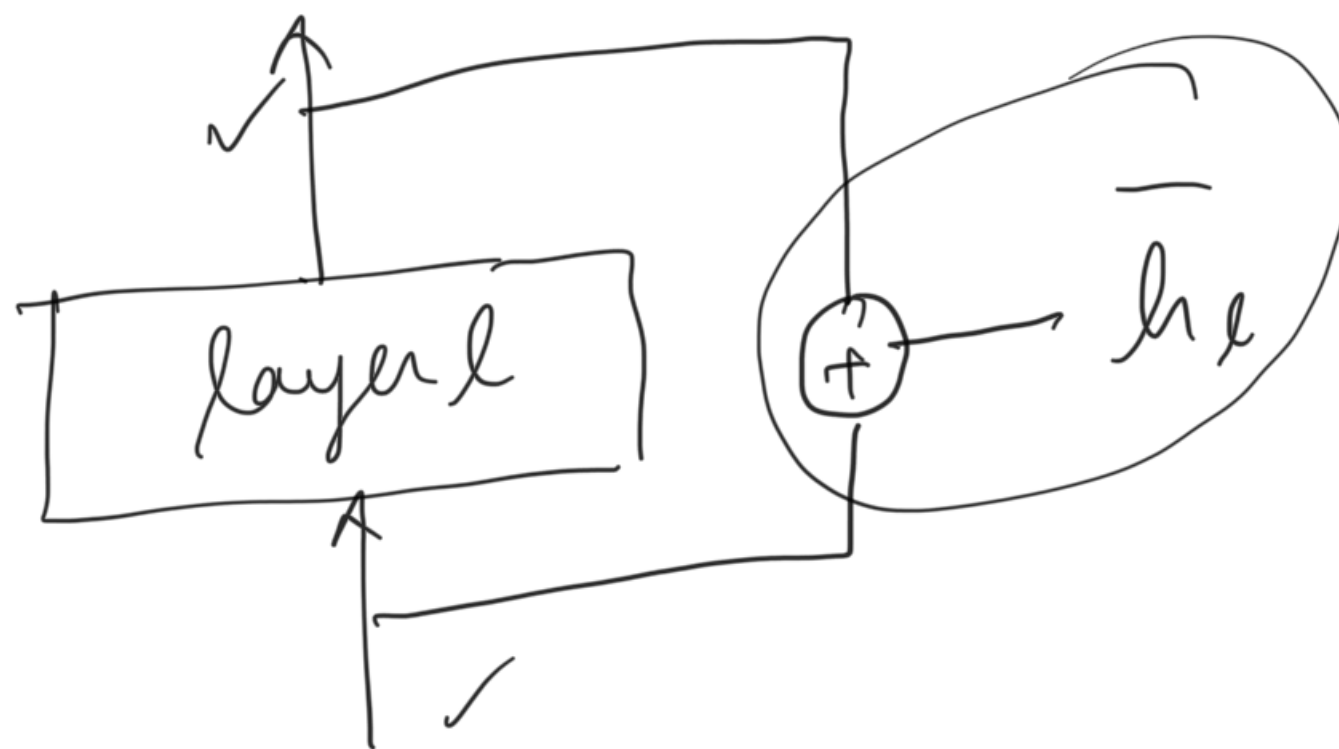
h_{12}^t (LPT-J).

→ predict the antonym with $\sim 24\%$ accuracy in a zero shot context

Last token.



$\bar{h}_l$



Function vectors.

$$1M : f(\phi)$$

Autoregressive $\hookrightarrow$ prompt.

output $\rightarrow$ next token distribution
or $f(p)$ over the
vocab V

we write $f(p)[y] \in [0, 1]$ predicted
probability of output $y \in V$ in
response to the prompt p .

f : L layers

Each layer $l \leq L$ has a hidden
representation : $h_l \in \mathbb{R}^d$
 $\hookrightarrow a_{li}$ attention
 J heads.

$$h_l = h_{l-1} + \underbrace{m_l^T}_{\substack{\downarrow \\ \text{MLP/feed forward} \\ \text{part.}}} \left(\sum_{j \leq J} \right)$$

a_{lj} : the projection of the output of the j^{th} attention head.

Attention heads & hidden vectors are part of the transformer input

$$a_{lj}(p), h_l(p).$$

.. .. hidden layer

Decoder maps the last hidden state
to the output distribution

$$D(h_L(p)) = f(p).$$

For a task $t \in \mathcal{T}$, where $\mathcal{T}$ is the
universe of ICL tasks we construct a
dataset P_t of in-context prompts

each look like $p_i^t \in P_t$

$p_i^t \rightarrow$ sequence of N input-output
examples (x, y) followed by a

query x_{iq} corresponding to
 target correct response y_{iq}
 not part of the
 prompt and
 should be predicted
 by the LM

We choose p_i^t in a
 way such that $f(p_i^t) \rightarrow y_{iq}$
 with the highest probability

$$p_i^t = \{ (x_{i1}, y_{i1}) \dots (x_{iN}, y_{iN}), x_{iq} \}$$

$\tilde{p}_i^t \rightarrow$ garbled version of the in-context prompt

$$\tilde{p}_i^t = \left\{ (x_{i1}, \tilde{y}_{i1}), \dots, (x_{in}, \tilde{y}_{in}), x_{i'q} \right\}$$

where $\tilde{y}_{i1}$ is not the correct answer for x_{i1} & so on for all the other examples.

$$\text{ICL } p_i^t \rightarrow f$$

task-conditioned attention:

mean

$$\rightarrow \bar{a}_{lj}^t = \frac{1}{|P_t|} \sum_{p_i^t \in P_t} a_{lj}^o(p_i^t)$$

Run the transformer again on $\tilde{p}_i^t$
(garbled prompt)

Causal Indirect Effect (CIE)

$$CIE(a_{lj} | \tilde{p}_i^t) = f(\tilde{p}_i^t | \bar{a}_{lj}^t) [y_{ij}^t] - f(\tilde{p}_i^t) [y_{ij}^t]$$

$\bar{a}_{lj}^t = a_{lj}^o$

← Causal impact of the

task - conditioned
attention.

↓ function vectors, (formal),

Average Indirect effect (AIE).

$$t \in \mathcal{T}$$

$$AIE(a_{xj}) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{1}{|P_t|} \sum_{\substack{p_i^t \in P_t}} CIE(a_{xj} | p_i^t)$$

↓
compute AIE

$\frac{1}{n} \sum_{i=1}^n x_i$

different heads(i) & different
layers(l)

16 heads & 32 layers

16×32 2512 A/E's

Sort the a_{ij} 's based on
their A/E values-

Gather the top a_{ij} from this
sorted list in a set A

1 1 1 1 1

$$|A| = 10$$

For a given task t , the function vector v_t is defined as

$$v_t = \sum_{a_{lj} \in A} a_{lj}$$

Model	Parameters	$ L $	$ A_L $
GPT-J	6B →	28	16
	20B →	44	64
GPT-NeoX	7B →	32	32
Llama-2.	13B →	40	40

70B → 80 64
Tasks (T) — 40 Simple tasks,

6 tasks.

Antonym:
Capitalization: Input word, output
first char in UpperCase

Country - Capital:
English - French:

Present - Past

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Singular - Pearson.

~~Input prompt:~~

Zero shot prompt $[x_{iq}]$

(a)

shuffled label prompt

(b) $[(x_{i1}, \tilde{y}_{i1}), (x_{i2}, \tilde{y}_{i2}) \dots (x_{iN}, \tilde{y}_{iN}) [x_{iq}]]$